Chapter Seven

Cities that Think

Designing the Future

Local governments spend lots of time and resources attempting to update our current cities, roads, and transportation systems. The cost of operation and maintenance and overall inefficiency is high. It is less expensive to build newer cities from the ground up than to restore and maintain old ones, just as it is more efficient and less costly to design flexible, state of the art production methods than it is to attempt to upgrade obsolete factories.

To have a world without pollution and waste, yet keep parks, playgrounds, art and music centers, schools, and health care available to everyone without a price tag, requires profound changes in the way we plan our cities as well as our lifestyles.

To phase in this new system, the first city will test the validity of design parameters and make necessary changes as needed. This new social direction could be promoted on many fronts with books, magazines, TV, radio, seminars, theater, and theme parks, We could also design and experiment with automated building processes for the next city.

Innovative multi-dimensional circular cities combine the most sophisticated resources and construction techniques available. The geometrically elegant circular arrangement, surrounded by parks and gardens, is designed to operate with minimum energy to obtain the highest possible standard of living for everyone. This city design uses the best of clean technology in harmony with local ecology.

The design and development of these new cities emphasizes the restoration and protection of the environment. It must be understood that technology without human concern is meaningless.

The new cities would provide a total environment with clean air and water, health care, good nutrition, entertainment, access to information, and education for all. There would be art and music centers, fully equipped machine shops, science labs, hobby and sports areas, and

manufacturing districts. These new cities would also provide all manner of recreation within a short distance of the residential district. Waste recycling, renewable and clean power generating systems, and all services would be managed with integrated, cybernated methods. The management of one's personal life, his/her life style and personal preferences, is left entirely to the individual.

Some cities can be circular while others may be linear, underground, or constructed as floating cities in the sea (we will get to them later on). Many cities would be designed as total enclosure systems, much like a cruise ship outfitted for a six-month cruise. They would contain residences, theaters, parks, recreation, entertainment centers, health care and educational facilities, and all the requirements and amenities for a total living environment. Everything in these cities would be as near to a self-contained system as conditions allow. In northern locations, some could be partially underground.

In planning the cities, computers will help determine the design based on the most comprehensive analysis of data about the environment and human needs. For example, the characteristics of the population in a given area determine how many hospitals and schools are built and the equipment needed. Some medical systems would be mobile and others prefabricated on land and sea. Eventually entire cities would be automatically assembled on site from standardized, prefabricated sections made in automated plants. Through this "systems approach" method -- we can't emphasize this too much -- we will be able to give all people a very high standard of living in the shortest possible time.

This permits a wide range of flexibility in design for changes and takes advantage of interchangeable units. Cities take on new and different appearances depending on how they are used. Each city is unique. It does not reduce the lives of people to a subsistence level; rather, it makes available all the amenities that modern science and technology can provide. Even the wealthiest people of the past could not achieve a standard of living equal to that in these new cities, which will also maximize safety and peace of mind.

The structures will be made of newer materials such as a sandwich-type assembly that is semi-flexible with an inner foam core and a glazed ceramic outer surface to allow for expansion and contraction without fracture. This requires no maintenance. The thin shell construction can be mass-produced in a matter of hours. This type of construction suffers little or no damage from earthquakes, hurricanes, termites, and fires. Windows will be controlled electronically to shade or darken external illumination

and come equipped with computer-controlled, automatic cleaning systems that require no human labor.

Innovative technologies make it possible to conserve resources for lesser-developed regions, without sacrificing any of the conveniences of advanced living. It is only through such innovations that our end goal of a high standard of living for the entire human race can be achieved.

These cities coordinate production and distribution, operating a balanced-load economy so there is no over or under production. Accomplishing this requires an autonomic nervous system (environmental sensors) integrated into all areas of the social complex.

For example, in the agricultural belt electronic probes embedded in the soil automatically maintain a constant inventory of the water table, soil conditions, nutrients, etc., and act appropriately without the need for human intervention as conditions change. This method of industrial electronic feedback would be applied to the entire system.

The cities would function as evolving, integrated organisms rather than as static structures because their design accommodates to change. These total environments will permit the widest possible range of individuality and creativity for those living in them.

Design Consideration

At one time, architectural adornments were an integral part of construction. The lofty columns and colonnaded porticos of ancient Greece and Rome were necessary components of their structures. With the advent of newer, lightweight materials and engineering improvements, we can now span greater distances without columns or other intervening support structures.

The resource-based economy would no longer engage in the conscious withdrawal of efficiency to maintain designs felt to be impressive. If we continue to design our buildings with conspicuous waste and decoration, we lessen the standard of living for others by using resources wastefully. Designing a building with many artificial projections does not indicate originality, creativity, or individuality. Individuality is expressed by our unique way of thinking about ourselves and the world around us, not by our external appearance.

This is not to detract from beautiful structures created in the past with the available and limited technology available at the time. However, the

continuing application of ancient methods of construction retards innovative and creative thinking, which is necessary to an emergent culture.

The intelligent use of resources incorporated into structures considerably simplifies our lifestyle and reduces waste and maintenance. These new cities would provide for the needs of the inhabitants through an efficient allocation of resources and materials, in an energy-conscious and pollution-free environment.

Homes

To many in the early twenty-first century, the homes of the future may appear surreal. For example, homes can be sheltered from the weather by electronic means. The furnishings may consist of totally different configurations that automatically adjust to our body contours. New technologies will make walls entirely transparent so occupants can view the surrounding landscape without anyone on the outside seeing in. Daylight can be softened and subdued according to the preference of the occupants. These buildings would provide a barrier to sound, insects, and dust, and maintain the desired internal temperature. Telephones would be entirely invisible and a component part of the interior structure, focusing sound to your ear by electronic means. The building's materials will generate energy and control their own surrounding climate.

With the intelligent application of humane technologies, a wide array of uniquely individual homes can be provided. Structural elements would be flexible and coherently arranged to best serve each individual. Pre-fabricated modular homes will embody a high degree of flexibility inconceivable in the past. They can be built in any place one might want, amidst forests, atop mountains, or on remote islands. They can be designed as self-contained residences with thermal generators, heat concentrators, and photovoltaic arrays built into the skin of the building. Thermo panes would tint out bright sunlight using variable patterns of shading. All these features are controlled by the occupant and supply more than enough energy to operate the entire household.

Homes will also contain a precise combination of dissimilar metals utilizing the thermocouple effect for heating and cooling. Other materials embedded in solid-state plastic or ceramic materials would make up the structure of the home. With this application, the warmer it gets on the outside, the cooler it becomes on the inside. This method serves to heat or cool the buildings. The interiors of the homes would be designed to suit the preferences of the individuals.

Transportation

When travel outside the city is desired, computer-guided vehicles for land, sea, air, space and beyond can transport passengers and freight. For rapid movement of passengers on land across viaducts, bridges, and tunnels, high-speed mag-lev trains span great distances and will efficiently replace most aircraft transportation. Some passenger compartments in the transport units can be transferred from the moving train during transit, which eliminates waiting time at stations. Rail, sea, and undersea craft can handle most freight. Many of the transport units have detachable components and contain standardized containers making them easy to transfer.

In the cities various types of escalators, elevators, conveyors, and transveyors can be designed to move in all directions throughout, even up the sides of buildings. They can be interconnected with other transport systems and extend into homes as well.

Most of the smaller transportation units for people can be operated by voice control. When voice control is not practical or possible, alternative methods such as keypads can be used. Without large corporations controlling automobile manufacture for profit, all transportation systems can be designed as modular, continuously updated, and provided with the latest developments in technology.

Circular City



The outer perimeter is part of the recreational area with golf courses, hiking and biking trails, and opportunities for water sports. A waterway surrounds the agricultural belt with its enclosed, transparent buildings. The application of newer technologies eliminates, once and for all, the use of dangerous chemicals and pesticides. Continuing into the city center, eight green sectors provide clean, renewable sources of energy using wind, thermal, and solar energy devices. The residential belt features beautiful landscaping, lakes, and winding streams. The homes and apartments are gracefully contoured to blend in with the landscape. A wide range of innovative architecture provides many choices for the occupants.

Adjacent to the residential district a wide selection of healthy, organically-grown foods are available on a 24-hour basis. Next are the apartments and design centers, which surround the central dome. Eight domes house the science, art, music, research, exhibition, entertainment, and conference centers, which are all fully equipped and available to everyone.

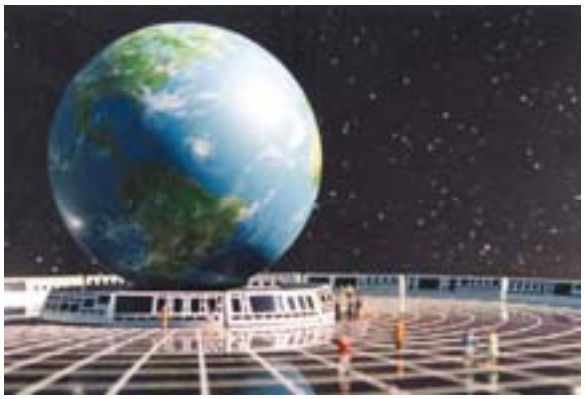


The central dome, or “theme center,” houses the cybernated system, educational facilities, health center, and facilities for shopping, communications, networking, and childcare. In addition, it serves as the core for most transportation services, which take the form of horizontal, vertical, radial, and circular conveyors that safely move passengers anywhere within the city. This system facilitates efficient transportation for city residents, eliminating the need for automobiles. City-to-city transportation is provided by monorail and electrically operated vehicles.



Total Enclosure Cities

Many cities are designed as total enclosure systems, much like a cruise ship outfitted for a six-month cruise. They contain residences, theaters, parks, recreation, entertainment centers, health care, and educational facilities, and all the requirements and amenities for a total living environment. Everything in these cities is as near to a self-contained system as conditions allow. In northern locations or uninhabitable areas, cities can be subterranean.



Cybernated Complex

This cybernated complex utilizes advanced imaging technology to project a 3-D "virtual" image of the earth in real time. It utilizes satellite communication systems to provide information on worldwide weather conditions, ocean currents, resource inventories, population, agricultural conditions, and fish and animal migration patterns. The interconnected cybernated complexes represent the brain and nervous system of the entire world civilization. All information is available on demand to anyone via the Internet. This single site manages our common heritage of resources, and monitors the carrying capacity and health of Earth.



University City

This University of Architecture and Environmental Studies, or "World University," is a testing ground for each phase of architectural development. This is a "living" and continually evolving research institute open to all. Student performance is based on "competence accreditation" and research findings are applied directly to the social structure to benefit all of humanity.

People live in these experimental cities and provide feedback on the livability and serviceability of the various structures. This information is used to formulate modifications to structures so that maximum efficiency, comfort, and safety are assured. This facility is also used to develop modular construction systems and components that serve a wide range of needs and preferences. In most instances, the external appearance of the buildings reflects the function of the building – they are designed "from the inside out."

Skyscrapers

These skyscrapers are constructed of carbon fiber-reinforced and pre-stressed concrete. They are stabilized against earthquakes and high winds by three massive, elongated, tapered columns, which are 100 feet wide at the base. This tripod-like structure is reinforced to diminish compression, tension, and torsion stresses.



These super-sized skyscrapers assure that more land is available for parks and wilderness preserves, while concurrently helping to eliminate urban sprawl. Each of these towers encloses a complete environment, containing access centers as well as childcare, educational, health, and recreational facilities. This helps reduce the need to travel to outside facilities.







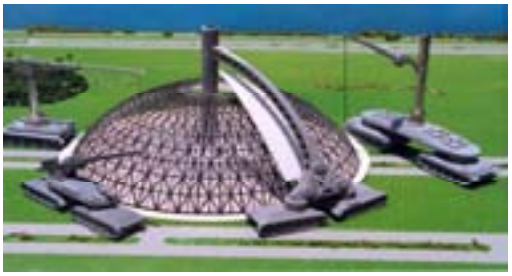






Center For Dialogue

The mission of The Center for Dialogue is to submit the urgent issues of the times to critical examination, and to raise relevant questions for informed public dialogue.



This photo on the right represents the automated building of a dome's structure.

International Shipping Systems





Hydrodynamic seafaring vessels permit high-speed efficient travel. They are energy efficient and provide maximum comfort and safety for passengers. They are manufactured from durable composite materials, their outermost skin consisting of a thin layer of titanium, which requires minimal maintenance. Portions of the upper deck slide open when the weather permits.

Ships With Detachable Components





Many of the transport units have detachable components and contain standardized containers making them easy to transfer. Entire freight sections are unloaded instead of separate containers.



Ships & Canals

An efficient national transportation system would include a network of waterways, canals, and irrigation systems. Many of the ships traversing these canals are floating automated plants while others transport passengers and freight. An innovative approach to education could be offered in floating "education centers" where children and adults travel from one area of the continent to the other. This would allow them to learn about the world in which they live, not merely through books, but through experiencing and interacting with the "real-world" environment.

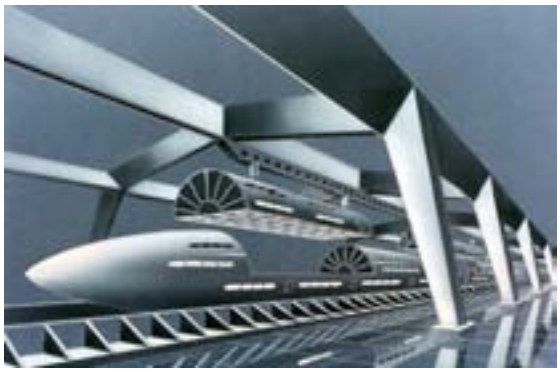
"Mega hydrological projects" would be an integrated part of intercontinental planning, minimizing floods and droughts while aiding the migration of fish, removal of accumulated silt, and developing sites to manage and "clean" agricultural and urban run-off. Floodwaters would be diverted to water storage basins, allowing the water to be used during periods of drought. This not only helps maintain the water table, but also provides natural firebreaks and an emergency water source for fires. In addition, these canals would supply water for farming and irrigation, supply the nation's land-based fish farms, protect the wetlands and wildlife, and supply water to recreation areas.

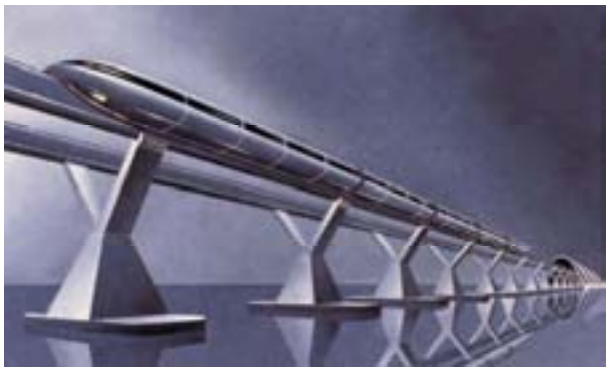


Automobiles

Streamlined cars would provide high-speed, energy efficient, and safe long-range transportation. Some vehicles may have wheels, while others are equipped with magnetic levitation or air-floatation devices. Vehicles would be equipped with voice-recognition technology that allows passengers to request their destination by voice command. Self-monitoring systems will tell the vehicles when service is required, and they would transport themselves to service and maintenance facilities. Use of clean, non-polluting electrical energy would allow for silent vehicle operation. Proximity-sensor devices linked to automated velocity and braking systems can enable the vehicles to avoid collisions. As a secondary safety measure, the entire interior will contain a protective membrane. Within the cities, horizontal, vertical, radial, and circular transveyors would serve most transportation needs.

Mag-Lev Trains - Mass Transportation Systems and Monorails





While these high-speed, magnetic levitation trains are in motion, a segment of the passenger compartment is either lifted or slid to the side.



These detachable sections can then take passengers to their local destinations while other compartments are lowered in their place. This method allows the main body of the train to remain in motion saving time and enhancing efficiency. In addition, the

removable compartments are specially equipped to provide a wide range of transportation services. These high-speed mag-lev trains and monorails will be used for city to city transportation.

Bridges





These elegant bridges are designed to carry compression, tension, and torsion loads in the simplified expression of its structural members. In some instances Mag-lev trains are suspended beneath the enclosed traffic lanes.

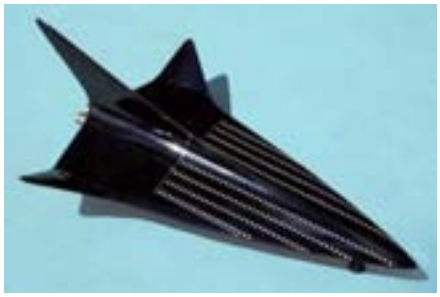




Future Aircraft

Since military aircraft would be unnecessary in a resource-based economy, emphasis is shifted to advancing medical, emergency service, and transportation vehicles. Here is an example of VTOL (Vertical Take-off and Landing) aircraft with three synchronous turbines, which allow exceptional maneuverability.





These delta-configuration aircraft are controlled by electrodynamic means, eliminating the need for ailerons, elevators, rudders, spoilers, flaps, or any other mechanical controls. In addition to providing better maneuverability and aerodynamic qualities, this innovative technology also serves as an anti-icing system. In the event of an emergency landing, fuel is ejected to prevent fires.



VTOL



Chopper



Chopper & Delta Wings

These Vertical Takeoff and Landing (VTOL) aircraft lift passengers and freight by the use of ring-vortex air columns. The helicopter in the foreground has a stationary center around which the rotors are propelled by engines at their tips. VTOL aircraft are propelled by a variety of techniques, from ducted fans to vectored jets. They are designed to combine the most desirable attributes of fixed winged aircraft, helicopters, and flying platforms. Transcontinental travel is achieved through advanced aircraft and high-speed mag-lev trains, all integrated into a worldwide transportation system.

Airports



The central dome of this airport contains terminals, maintenance facilities, service centers, and hotels. The runways are arranged in a radial configuration, which allows airplanes to easily take off into the prevailing winds and to avoid dangerous crosswind landings. Emergency stations at the edge of the runways are fully equipped with built-in fire fighting equipment and emergency

arresting gear. All runways are equipped with built-in sprinkler systems. Passengers are transported to and from the airport by underground conveyors. Many of the terminals themselves are constructed underground for increased safety and more efficient use of the land.





Homes

The architecture and individual dwellings of the cities evolve in an entirely different manner from past houses. With the intelligent application of humane technologies, a wide array of unique individual homes can be provided. Their structural elements are flexible and coherently arranged to best suit individual preference. These pre-fabricated modular homes, embodying a high degree of flexibility, can be built in any place one might imagine, amid forests, atop mountains, or on remote islands.







All these dwellings are self-contained energy efficient residences with their own thermal generators and heat concentrators. Photovoltaic arrays are built into the skin of the building and into the widows. “Thermopanes,” tint out bright sunlight with variable patterns of shading. All these features are selected by the occupant to supply more than enough energy to operate the entire household.








Homes are prefabricated from a new type of pre-stressed reinforced concrete with a flexible ceramic external coating; they are relatively maintenance-free, fireproof, and impervious to weather. Their thin shell construction is mass-produced in a matter of hours. With this type of construction, there is minimal damage from earthquakes and hurricanes.






Automation

Machine Intelligence

The key to achieving abundance and a high standard of living for every person on the planet is to automate as much as possible in the shortest period of time. By replacing human labor with machines and implementing a global resource-based economy, everyone lives better than the wealthiest of today. A future without locks on the doors, and no fear of someone hitting you over the head to steal your goods or money is possible -- because everyone has free access to all that is available. And there can be much available by automating and using resources wisely.

Cybernation, the merging of computers with production, will unleash an outflow of goods and services never before experienced. There is much work to be done on artificial intelligence (AI). AI is computer programming that simulates human decision-making and hypothesis testing along with self-correction. AI redesigns mechanical and electronic systems to simulate and improve upon human performance. As exciting as these developments are, we are just at the beginning of the possibilities.

Cybernation can be regarded as the only *real* emancipation proclamation for humankind when used humanely and intelligently. It enables people to have the highest conceivable standard of living with practically no labor. It frees people for the first time from the routine of repetitive day-by-day activity. When humans are free of the outmoded methods of a monetary system, we will finally begin to understand what it means to be civilized.

When we use automation and cybernation more extensively, not only industrial workers, but also most professionals can be replaced by machines. Even today, the most visionary writers and futurists have difficulty accepting the possibility of robots replacing surgeons, engineers, top management, airline pilots, and other professionals. Machines can easily replace humans in government and in the management of world affairs. This does not represent a take-over by machines, as some people might fear. Instead, the gradual transfer of decision-making to machine intelligence is the next phase of social evolution.

Computerized systems are more effective because of the number of sensors they have. Automated control can happen when sensors are installed in every conceivable location and linked through a worldwide network of computers.

During the transition from a monetary society to a resource-based society, teams of systems engineers, computer programmers, systems analysts, researchers, and the like will be needed to help supervise, manage, and analyze the flow of goods and services. But as the resource-based society moves toward a more cybernated world, most people are no longer needed to manage and operate this emerging civilization. Computers will eventually be able to design their own programs, improve and repair their own circuitry, and update information about social needs. Interlinked cyber-centers would coordinate the service industries, transportation systems, public health care, and education with the latest data for the world economy. Redundant systems would be in place in the event of failures or interruptions.

AI organized in this way within a resource-based economy results in more significant changes for the human race than any previous breakthrough, philosophy, or revolution. It is most significant when we understand that what is blatantly missing in today's society is the intelligent management of Earth's resources, and that most problems can be solved when technology is used wisely.

Consequently, a much higher standard of living for everyone all over the world can be achieved when the entire Earth's resources are connected, organized, monitored, and used efficiently for *everyone's* benefit as a total global system -- not just for a relatively small number of people.

Mega-machines

Mega-machines represent radical changes in the physical appearance, performance, and behavior of machines. They act more like living systems since they are capable of making appropriate decisions for their particular uses. In the event of unforeseen threats or dangers to humans, they will act on our behalf. To minimize system failure, computers are designed to be flexible and are capable of automatic shutdown in the event of failure of one of their parts.

In an advanced cybernated global economy, mega-machines, directed by sophisticated AI, excavate canals, dig tunnels, and construct bridges,

viaducts, and dams without the need for human involvement. Human participation consists of selecting the desired ends.
Massive self-erecting structures are most efficient in the construction of the entire global infrastructure. These are not cookie-cutter cities, as some would imagine. The notion that large-scale overall planning implies mass uniformity is incorrect.

Factories can be designed by robots for robots; cybernated systems are self-programming by means of environmental feedback. Machines of the future are capable of self-replication and improvement, and can repair themselves and update their own circuitry. Since the computers and systems involved are self-monitoring, parts are supplied and installed well in advance of any wear. The machines operate continuously except when conducting their own maintenance and repair.

To conserve energy, which is a major theme in the resource-based society, many manufacturing plants can be eliminated by having products produced automatically while being delivered. For example, transportation modules for ships, trains, and planes can process time-sensitive products like fish and vegetables while en route.

Utilizing technology in this way makes it possible for a global society to advance and change in the shortest time possible. You should remember that this is all made possible because the main objective is for everyone's gain, not just the benefit of certain corporations or individuals.

Robots Constructing Robots -Multi Access Industrial Robots



These "multi access industrial robots" utilize vast information resources, which enable them to receive commands via satellite up-link or on-site. They are also designed to take appropriate actions in the absence of human directives by combining an array of Micro Electro Mechanical Systems (MEMS), sensors, and receivers with sophisticated decision-making circuits and artificial intelligence programs. They are capable of handling a wide range of industrial production tasks, and are even able to upgrade their level of service and replace their own parts. When necessary, these mega robots communicate with one another and coordinate the logistics and delivery of the required materials for each project.

Nanotechnology

Nanotechnology offers enormous potential. Nanotechnology combines optics and lasers, and enables matter to be assembled atom by atom, into whatever molecular structure is needed. Nanotechnology will lead to a sub-microscopic revolution in all fields.





Mega-Excavation Machines

This scene depicts a laser excavator. Such devices, directed via satellite, are capable of fusing the earth beneath it into a molten, magma-like material, thus contouring the earth to aid in the construction of canals, roads, and waterways.

Automated Tunnel Assembling Machine

Tunnel segments float down canals by the use of large flotation devices. This automated tunnel-assembling machine lifts prefabricated segments and places them in the required positions. Once completed the tunnels are used for high-speed mag-lev transportation.



The Construction of Towers



These towers are designed specifically for regions where earthquakes are prevalent. Such cable-suspended structures easily withstand a wide range of movement, stresses, and strains. Circular towers radially arranged are self-erected efficiently and rapidly around a central core that houses elevators and all other household utilities. The translucent windows serve as photovoltaic generators, and the intensity of the light entering the windows is varied electronically. All window cleaning and maintenance is automated.

Massive Lifting Crane



This multi-function crane is designed to lift freeform structures and position them on foundations and stilts or transfer them to vertical lift systems that place the units on towers. Upon completion of their task, these self-erecting cranes are disassembled into a compact form for easy transportation to the next assignment.

Mass-Produced Extruded Dwellings

This demonstrates how lightweight carbon fiber-reinforced concrete apartment dwellings are produced as continuous extrusions and then separated. The outer shells of these efficient structures serve as photovoltaic generators.





Lift and Positioning Crane

This automated machine places these prefabricated dwellings at on-site locations.

Mega Machines

The machine in the foreground is a multi-function unit. As shown, it lifts and inserts prefabricated housing components into a support structure. Although these "prefab" units are composed of standard components,



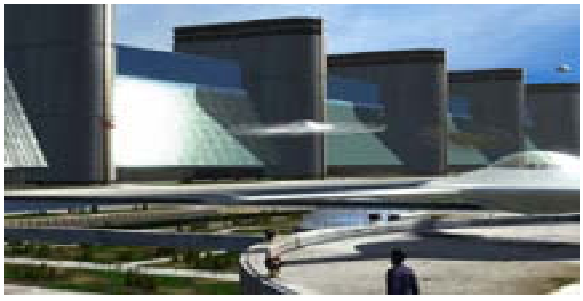


they are a modular design of such diversity as to allow maximum individual expression in interior design and décor.

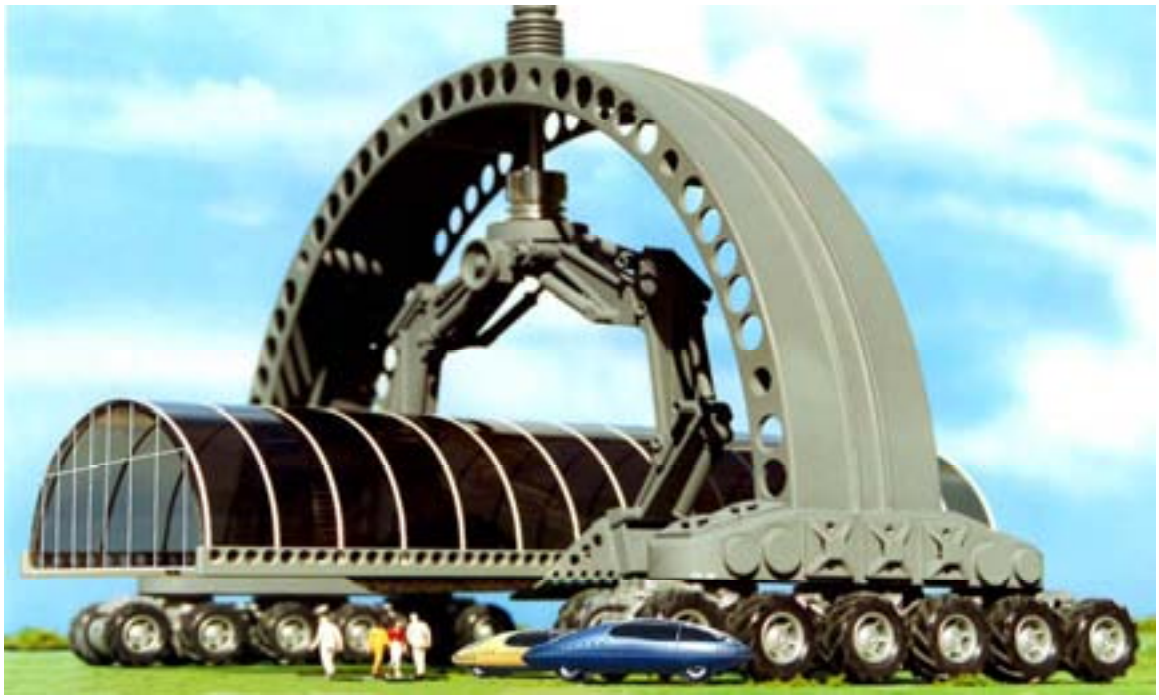


Automated Construction Systems

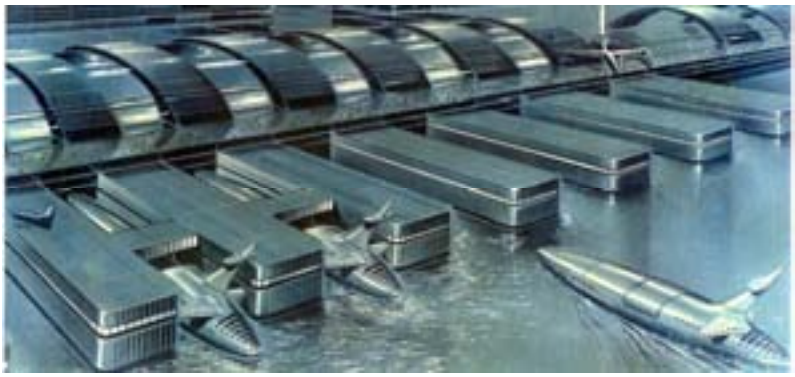
Construction of these industrial and research complexes is carried out by robotic equipment that receives instructions via satellite. The construction crews consist of automated cranes that travel along the buildings' length installing floors, windows, curtain-walls, roofing, and other components from the ground up, entirely free from human intervention. These devices contain self-monitoring sensors to minimize industrial accidents or collisions with other devices or living beings.



Desalinization Plant



This mega-machine is transporting a transparent enclosure used for evaporative condensation. It is placed over canals, some of them containing salt water, and serves as an evaporative desalinization plant to supply clean water for drinking, irrigation, and other needs. This is accomplished by harnessing the power of the sun and helps eliminate water shortages throughout the world.



International Shipping Systems

These ships are floating automated plants, capable of processing raw materials into finished products while en route to their destinations. Some serve as industrial fish processing plants and canneries, while others are equipped with multi-cellular compartments capable of transporting a wide variety of products.